DOCUMENTATION

Information Security Project Proposal

DOCUMENTATION :

Multi-Layer Canary Token Based Intrusion Detection and Alert System

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1. Introduction

Information security is one of the most important areas in modern computing systems. With the increasing number of cyber-attacks, unauthorized access to sensitive files has become a major concern. Traditional security systems mainly focus on preventing external attacks, but they often fail to detect insider threats or suspicious file activities inside a system.

This project presents a simple simulation of an intrusion detection system using canary tokens (decoy files). These decoy files are used to detect unauthorized or suspicious access attempts and generate alerts in real-time.

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2. Problem Statement

Attackers or malicious users may try to access sensitive files such as passwords, configuration files, or secret keys. While traditional security mechanisms can restrict login access, they may not detect internal file-level activities.

There is a need for a lightweight system that can:

- Detect unauthorized access attempts
- Monitor file-level activity
- Generate alerts when suspicious actions occur
- Maintain logs for analysis

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3. Objectives

The main objectives of this project are:

- To create decoy (canary) files that act as traps for attackers
- To monitor file access, modification, and deletion
- To generate real-time alerts on suspicious activity
- To maintain logs of all events with timestamps
- To demonstrate a basic deception-based security system

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4. Scope of the Project

This project is a simulation-based implementation of an intrusion detection system. It is designed for educational purposes and demonstrates how canary tokens can be used for security monitoring.

Scope includes:

- Creation of fake sensitive files (canary files)
- Detection of file operations (access, modify, delete)
- Alert generation with sound and visual indicators
- Logging system with timestamps
- Admin-based log viewing system

Limitations:

- It does not provide real network-level security
- It does not use machine learning or AI-based detection
- It is limited to local system file simulation only

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5. System Design

The system works in the following way:

1. Canary files are created at the start of the program
2. User is given a menu to perform file operations
3. If a file is accessed, modified, or deleted:
    - An alert is generated
    - Sound notification is played
    - Activity is logged with timestamp
4. Admin can view logs using a password-protected option

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6. Tools and Technologies Used

- Programming Language: C++
- IDE: CodeBlocks / Dev C++ / Visual Studio Code
- Libraries Used:
    - iostream (input/output handling)
    - fstream (file handling)
    - ctime (timestamp generation)
    - windows.h (alerts and sound)

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7. Features of the System

- Creation of decoy files (canary tokens)
- File access monitoring
- Real-time alert system
- Sound-based notification using MessageBeep
- Logging system with timestamps
- Admin-protected log viewing
- Input validation to avoid system errors

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8. Working Methodology

The system starts by creating fake sensitive files such as passwords, configuration data, and secret keys. These files act as traps for unauthorized users.

When a user interacts with the system, they are provided with options to access, modify, or delete files. If any action is performed on a monitored file, the system immediately:

- Displays a security alert on the screen
- Plays a sound notification
- Saves the event in a log file with timestamp

An admin can later view all logs by entering a secure password.

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9. Conclusion

This project demonstrates a simple implementation of a deception-based intrusion detection system using canary tokens. It shows how fake files can be used to detect unauthorized access attempts.

Although this system is a simulation and not suitable for enterprise-level deployment, it effectively demonstrates the core concept of intrusion detection and security monitoring.

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